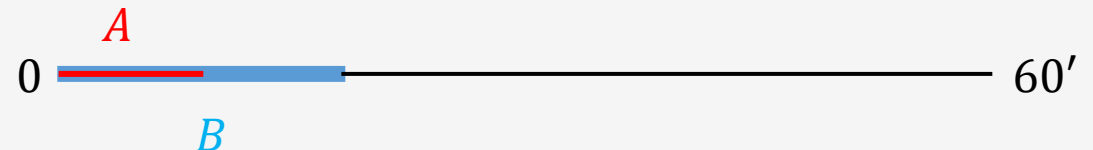

Conditional Probability

Conditional Probability

- **Let's continue with Example 2:**
 - We wait at the bus stop and the bus comes in the first 60'
 - **Modeling assumption:** The bus is equally likely to arrive at each time
 - Let's define **two events**
 - A = "the bus arrives in the first 10 minutes" $\rightarrow A = [0, 10']$
 - B = "the bus arrives in the first 20 minutes" $\rightarrow B = [0, 20']$
 - **What is $P\{A\}$ = ??**



Conditional Probability

- **Let's continue with Example 2:**

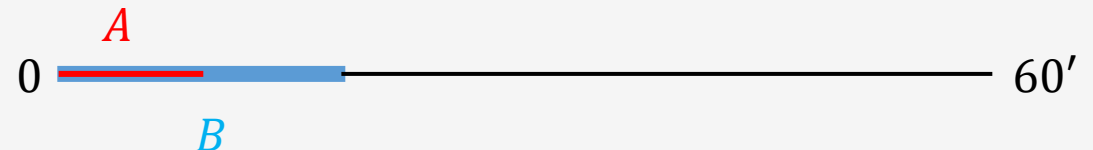
- We wait at the bus stop and the bus comes in the first 60' (1 hour)
- **Modeling assumption:** The bus is equally likely to arrive at each time

- Let's define **two events**

- A = "the bus arrives in the first 10 minutes" $\rightarrow A = [0, 10']$
- B = "the bus arrives in the first 20 minutes" $\rightarrow B = [0, 20']$

- **What is $P\{A\}$ = ??**

- **If you know $B = \text{TRUE}$ what is $P\{A\}$ = ??**



Conditional Probability

- **Definition:** Suppose you have two events A, B . The conditional probability of event A given that the event B happened (i.e., $B = \text{TRUE}$) is defined as

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

- **Intuition:**
 - Run your random experiment many times and record the results for events A **and** B (you have $A = \text{True/False}$ and $B = \text{True/False}$);
 - **Keep only** experiments where B happened (you now have $A = \text{True/False}$ and $B = \text{True}$);
 - Look at the **fraction** in these experiments that A happened too.

Conditional Probability

- **Example:**

- You have a die and consider the two events $A = \{2\}$ and $B = \{2,4,6\}$.
- What is $P(A|B) = ?$

- **Intuition:**

- Perform the random experiment many times and record outcomes
- Keep the experiments where $B = \{2,4,6\}$; ($B = \text{TRUE}$ and A can be anything)
- Find the **fraction** where $A = \text{TRUE}$

Conditional Probability

- **Is $P(\cdot | B)$ a probability??**

Conditional Probability

- Is $P(\cdot | B)$ a probability?? → Check if the 3 Axioms hold??

Conditional Probability

- Is $P(\cdot | B)$ a probability?? → Check if the 3 Axioms hold??

1. $0 \leq P(A|B) = \frac{P(A \cap B)}{P(B)} \leq 1,$ Why?

2. $P(\Omega|B) = \frac{P(\Omega \cap B)}{P(B)} = ??$

3. If A, A' are 2 mutually exclusive events ($A \cap A' = \emptyset$), then

$$P(A \cup A'|B) = ??$$

Conditional Probability

- **Is $P(\cdot | B)$ a probability??** → Check if the **3 Axioms** hold??

1. $0 \leq P(A|B) = \frac{P(A \cap B)}{P(B)} \leq 1$, because $P(A \cap B) \leq P(B)$ (by simple logic and $A \cap B \subseteq B$) implies

2. $P(\Omega|B) = \frac{P(\Omega \cap B)}{P(B)} =$??

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Conditional Probability

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Conditional Probability

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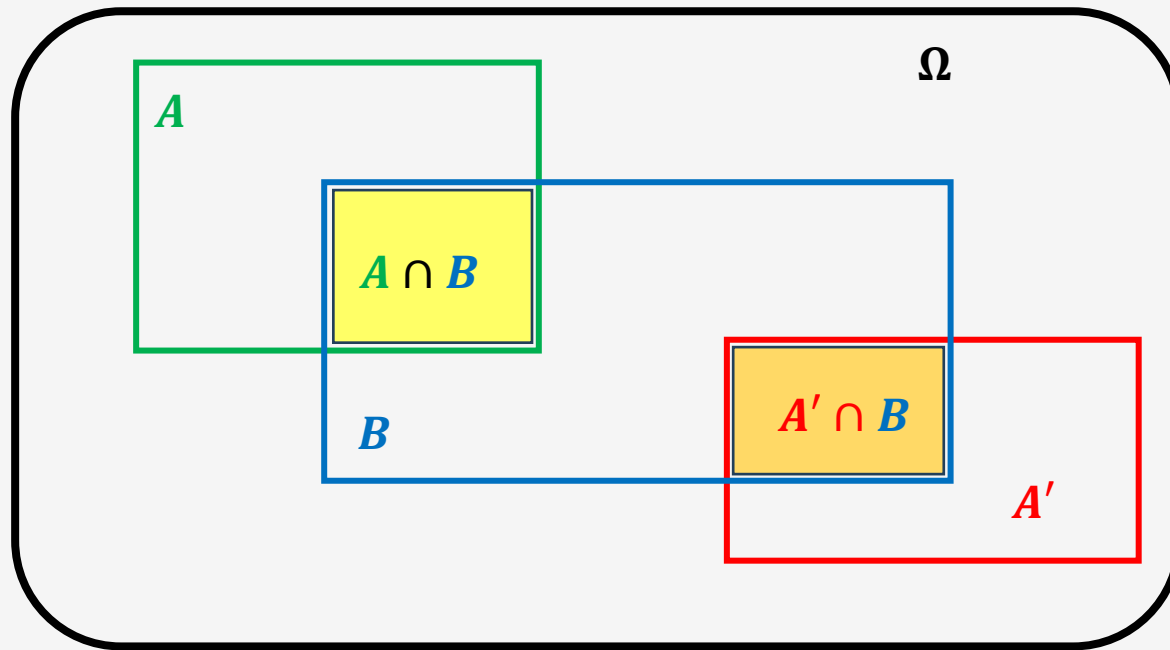
$$\begin{aligned} P(A \cup A'|B) &= \frac{P((A \cup A') \cap B)}{P(B)} = \frac{P((A \cap B) \cup (A' \cap B))}{P(B)} = \frac{P(A \cap B) + P(A' \cap B)}{P(B)} \\ &= P(A|B) + P(A'|B) \end{aligned}$$

Conditional Probability

- **Illustration of 3rd axiom**

$$P(A \cup A'|B) = P(A|B) + P(A'|B)$$

$$A \cap A' = \emptyset$$



Bayes formula

Bayes formula



Rev. Thomas Bayes
1701-1761

- **Connects** $P(A|B)$ **with** $P(B|A)$
 - **Useful when one of them is easier to compute than the other**
- **Example**: A popular COVID test has the following characteristics:
 - If a person is **sick**, the probability of testing positive is = 95%
 - If a person is **not sick**, the probability of testing positive = 2%
 - 1 person in 1000 is sick
- **Question**: What is the probability that a person is sick, when the test of that person comes back positive

Bayes formula

- What are our **Events??**
 - $S = \{sick\}$ and $Q = \{test\ is\ positive\}$
 - What probabilities we know already??
 - $P(S) = \frac{1}{1000}$, $P(Q|S) = 0.95$, $P(Q|S^c) = 0.02$
- Which probability we are looking for??
 - $P(S|Q) = ??$

Bayes formula



- We will use **Bayes formula** to compute $P(S|Q)$

$$\begin{aligned} P(S|Q) &= \frac{P(S \cap Q)}{P(Q)} \Leftrightarrow P(Q)P(S|Q) = P(S \cap Q) \Leftrightarrow \frac{P(Q)P(S|Q)}{P(S)} = \frac{P(S \cap Q)}{P(S)} \Leftrightarrow \\ &\Leftrightarrow \frac{P(Q)P(S|Q)}{P(S)} = P(Q|S) \Leftrightarrow P(S|Q) = \frac{P(S)P(Q|S)}{P(Q)} \end{aligned} \quad (1)$$

- In the above formula **we know**: $P(S)$ and $P(Q|S)$.
- We only **need to compute** $P(Q)$

Bayes formula

$\Omega = \{\text{all people}\}$

- But I can write: $P(Q) = P\left(Q \cap \overbrace{(S \cup S^c)}^{\Omega = \{\text{all people}\}}\right) = P((Q \cap S) \cup (Q \cap S^c)) =$
 $= P(Q \cap S) + P(Q \cap S^c) = P(Q|S)P(S) + P(Q|S^c)P(S^c) \quad (2)$
- From (1) and (2) we have **Bayes formula**:

$$P(S|Q) = \frac{P(S)P(Q|S)}{P(Q|S)P(S) + P(Q|S^c)P(S^c)}$$

- Substituting what we know yields: $P(S|Q) \approx 4.5\%$

Bayes formula

- This means that **only approximately 4.5% of the people who take the test and got a positive result are actually sick.**
- The test seems to be **USELESS!**
- Why is this happening?? The test has “impressive credentials”: $P(Q|S) = 0.95$ and $P(Q|S^c) = 0.02$
- **The Reason:** the disease is rare.
 - **The disease is less likely to happen, $P(S) = 0.001$, than the failure of the test $P(Q|S^c) = 0.02$.**

Indicator function & Expectation

Indicator function and Expectations

- **Example:**

- Random Experiment: Flip n **identical** coins. You are interested in the **total number of Heads** that may come up, **on average**, if you repeat the experiment many times.
- Define the DRV: $X = \text{"the total \# of H"}$
- **Question**: What is $E[X] = ??$
 - Define the **events**: $A_i = \text{"the } i\text{-th coin shows H"}$, $i = 1, 2, \dots, n$
 - $P\{A_i\} = p$ and $P\{A_i^c\} = 1 - p$

Indicator function and Expectations

- **Example (cont.):**

- Define the indicator function: $1_{A_i} = \begin{cases} 1, & \text{if } A_i = \text{True} \\ 0, & \text{otherwise} \end{cases}$

- Then we can write $X = 1_{A_1} + 1_{A_2} + \cdots + 1_{A_n}$

- We have: $E[X] = E[1_{A_1} + 1_{A_2} + \cdots + 1_{A_n}] = E[1_{A_1}] + E[1_{A_2}] + \cdots + E[1_{A_n}] =$
 $= P\{A_1\} + P\{A_2\} + \cdots + P\{A_n\} = np$

Indicator function and Expectations

- **Example (cont.):**

- **Note:** we did **NOT** have to assume that $A_i \perp\!\!\!\perp A_j, \forall i \neq j$. This means that even if we were **not** flipping the coins independently (but through a **collusion or conspiracy**) we would have gotten the same result: $E[X] = np$
 - A possible **collusion** or **conspiracy**: "I flip my coin and then I force everyone to turn their coins to my outcome".
- **Note:** We will often compute expectations using **tricks** (i.e., using indicator functions) similar with those in this example.

Conditional Expectations

Conditional Expectation

- **Definition:** If $X: \Omega \rightarrow D_X$ and $Y: \Omega \rightarrow D_Y$ are DRVs and $\varphi: D \rightarrow R$ then the **conditional expectation** of $\varphi(X)$ given $Y = y$ is

$$E[\varphi(X) \mid Y = y] = \sum_{x \in D_x} \varphi(x) P\{X = x \mid Y = y\}$$